Car Costing & Import File Cost Voucher

Training Manual — theory, workflow, and a button-by-button guide so anyone can understand and operate the system.

Audience: Operations, Accounts, and Super Users · **App:** vsd_jan_integration · **Site example:** N.TAN039/26MSC (BL MEDUW3445865) · **Excel source:** 1 BL COST.xlsx → sheet NEW COST FILE

> **How to use this manual:** Sections 1–4 explain the why and the model. Sections 5–7 are desk walkthroughs (every field and button). Section 8 is a full story with numbers. Section 9 is FAQ / mistakes.

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1. The business problem (in plain language)

Jan International imports cars from Japan. Each car ends up with many cost pieces: purchase price in Yen, Japan-side charges (inspection, shipping, ICD…), and Tanzania-side charges (TRA duty, wharfage, TPA, clearing, service…).

Those costs do **not** all arrive at the same time:

- Early: you know chassis, purchase Yen, later M3, later TRA duty assessment.
- Later: accounts posts real Journal Entries / Purchase Invoices against an **Import File** (supplier invoice no. like `N.TAN039/26MSC`).
- Some invoices are for **one car**; some are for the **whole Bill of Lading (BL)** and must be shared.

The costing module exists so the team can:

1. **Estimate** what each car should cost (for pricing / readiness) before every invoice is in.
2. **Allocate real posted expenses** onto the right cars when accounting documents exist.
3. Keep estimates and actuals connected through one shared dictionary of cost lines (**Costing masters**).



Figure A — End-to-end journey from imported cars to updated per-car costs.

2. Core idea: Projected vs Actual

Projected (Expected)

What we think the car will cost, based on:

- Fixed defaults (Ex-Bond, Service…)
- User-entered variables (TRA Duty, M3-driven shares)
- Formulas (Wharfage = Duty × 7%)

Where: Car Costing document.

Actual (Posted)

What accounts really booked in ERPNext:

- Journal Entry debit lines
- Purchase Invoice item amounts
- Linked to an Import File

Where collected: Import File Cost Voucher (IFCV).

> Your expense-list spreadsheet already said this: some lines are **Expected By Vehicle** but **Actual By BL** (e.g. Clearing Container, Wharfage, TPA). That gap is exactly why IFCV exists — it takes a BL total and divides it back onto cars.

Expense	SOURCE			Expected	Actual		
THC	MKT						
Forwarding	MKT						
Freight	MKT						
Inspection	MKT						
JPN Expense	MKT						
Repair Cost JPN	MKT						
Car Expense	MKT						
Vanning	MKT						
Transportation	MKT						
Freight & Other	MKT						
TRA DUTY	Expense List	VARIABLE		By Vehicle	By Vehicle		
CLEARING CONTAINER	Expense List	FIX		By Vehicle	By BL		
EX-BOND	Expense List	FIX		By Vehicle	By Vehicle		
SERVICE	Expense List	FIX		By Vehicle	By Vehicle	Car Service	
WHARFAGE	Expense List	VARIABLE	Container	By Vehicle	By BL	Duty * 7.5%	
DRIVER PORT TO YARD	Expense List	FIX		By Vehicle	By Vehicle		
FUEL PORT TO YARD	Expense List	FIX		By Vehicle	By Vehicle		
TPA	Expense List	VARIABLE	RORO	By Vehicle	By BL	Duty * 7.5%	
BOND INSURANCE	Expense List	FIX		By Vehicle	By Vehicle	Paid yearly and divided by Vehicle	
REPAIRING CHARGES Local	Expense List	VARIABLE		By Vehicle	By Vehicle		
JAPAN COST	Expense List	FIX					
ICD	Expense List	AS PER BL			By BL		
SHIPING LINE	Expense List	BL SHIPPING LINE DIVIDED ON ALL BL CARS M3			By BL		

Figure B — Business rules sheet: Expected vs Actual, Fixed vs Variable, By Vehicle vs By BL.

3. The three building blocks

DocType	One record means…	Main job
Costing	One type of cost (e.g. TRA Duty)	Master rule: Fixed/Variable/Formula, currency, GL account, By Vehicle vs By BL
Car Costing	One car on one Import File	Build projected TZ + Japan lines and landed COST
Import File Cost Voucher	One Import File / shipment batch	Pull posted JE/PI, allocate to cars, optionally push into Car Costings

Supporting idea: every car is an **Item**. Cars on a shipment share an **Import File** (named like the supplier invoice). Matching rule for vehicles:

1. Prefer `Import File.bl_number` → `Item.custom_bl`
2. If BL blank, use `Import File.supplier_invoice_no` → `Item.custom_bl`

4. Glossary

Term	Meaning
BL	Bill of Lading — shipping document grouping cars (e.g. MEDUW3445865)
Import File	ERPNext document for a supplier invoice / shipment batch (e.g. N.TAN039/26MSC)
M3	Cubic metres of the car — used to share shipping/ICD fairly
Costing	Named cost type with rules and optional GL account
Projected amount	Estimate on a Car Costing line
Allocation	Share of an IFCV expense assigned to one Item/car
By BL Equal	Split expense equally across all cars on the Import File
By BL M3	Split expense proportional to each car’s M3
By Vehicle Manual	User (or auto chassis match) assigns amounts per car
Landed COST	TOTAL COST USD × TZS/USD exchange (our projected COST)

5. Costing — every field explained

Open: Desk → **Costing**. This is configuration, usually maintained by a super user, not changed daily.

Field	What it does	How to explain it
Costing Code	Internal unique key (e.g. `tra_duty`)	Used by formulas and IFCV mapping. Do not rename casually.
Costing Name	Label people see (e.g. TRA DUTY)	Matches the expense-list wording.
Region	Tanzania or Japan	Controls which table on Car Costing the line appears in (TZS vs USD).
Entry Mode	Fixed Variable Formula	**Fixed** = system fills default (editable). **Variable** = user types the estimate for this car. **Formula** = system calculates (user should not invent the number).
Allocation Scope	By Vehicle or By BL	Tells IFCV how actuals usually arrive. By Vehicle → Manual mapping dialog. By BL → Equal or M3 split of one bulk amount.
Account	Link to Chart of Accounts	**Critical for IFCV.** When a JE/PI posts to this account, IFCV auto-selects this Costing.
Currency	TZS or USD (etc.)	Drives currency on Car Costing lines.
Default Projected Amount	Used when Entry Mode = Fixed	Example: Ex-Bond 139,314; Japan Cost 261; Inspection 170.
Display Order	Sort order on the form	Cosmetic / evaluation order only. Not Excel import sequence.
Formula / Depends On	Calculation expression	Example Wharfage: `{line.tra_duty} * 0.07` (matches Excel Duty×7%).
Disabled	Hide obsolete heads	Old heads (e.g. M3-as-cost-line) stay disabled, not deleted.

> **Teaching tip:** “Costing is the dictionary. Car Costing writes sentences with that dictionary. IFCV finds real invoices and attaches the right dictionary word via the Account link.”

6. Car Costing — every field & button

Open: Desk → **Car Costing**. One document = one car's projected cost worksheet.

6.1 Header fields

Field	What happens / what user does
Series	Auto name like `CC-2026-00024`. Leave default.
Item	**Required.** Pick the car Item. On a new doc, selecting Item triggers Load from Item.
Import File	Links this costing to the shipment (e.g. N.TAN039/26MSC). Auto-resolved from Item BL when possible.
Costing Date	Date of this worksheet (defaults Today).
Exchange Rate (TZS/USD)	**Required.** Converts USD summaries to TZS COST. Excel sample uses 2700.
YEN/USD Rate	Converts purchase Yen to USD (default 155).
Chassis / Maker / Model	Read-only, fetched from Item. For identification only.
M3	Car volume. Header field (not a cost line). Needed for M3-based shares and readiness.
YEN	Purchase price from Item (`custom_price`), read-only on form.

6.2 Child tables

Tanzania Costing Lines TZS

Examples: TRA Duty, Clearing, Ex-Bond, Service, Wharfage, Driver, Fuel, TPA, Bond Insurance, Local Repair.

Each row: Costing, code, mode, account, currency, **Projected Amount**, override flag, remarks.

Japan Costing Lines USD

Examples: Japan Cost, Inspection, ICD, Shipping Line.

Same structure as TZ table, different currency/region.

6.3 Summary fields (calculated — usually read-only)

Field	Meaning (Excel-aligned)
TOTAL EXP TSH	Sum of TZ lines **excluding TRA Duty** (Excel col Y = Q…X).
TSH TO USD	TOTAL EXP TSH ÷ Exchange Rate.
TOTAL EXP USD	TSH TO USD + all Japan line amounts.
YEN IN USD	YEN ÷ YEN/USD rate.
TOTAL COST USD	Duty USD + TOTAL EXP USD + YEN IN USD.

COST	TOTAL COST USD × Exchange Rate = **landed cost in TZS**.
PRICE	Currently same as COST (selling margin/VAT from Excel not applied yet).
Days Since Purchase	Informational age of stock.
Readiness	HTML checklist (Item, Import File, FX, M3, TRA Duty). Display only.

> Excel's COST column also adds City Levy + VAT derived from a selling price. Our COST is the **landed** figure (TOTAL COST USD × FX). Explain that difference when comparing to Excel.

6.4 Buttons on Car Costing

Control	When available	What it does
Load from Item	Always on form	Rebuilds TZ + Japan lines from active Costing masters; pulls chassis/ maker/model/Yen/M3 from Item; fills Fixed defaults; keeps existing projected amounts where present; recalculates; saves. On new docs, selecting Item calls this automatically.
Actions → Recalculate	Saved documents only	Re-runs Fixed + Formula lines and summary math. Shows alert with Cost / Price. Use after changing Duty, M3, rates, or overrides.
Save	Draft	Validates exchange rate and recalculates during validate.
Submit	Draft, when ready	Locks the Car Costing and copies projected lines + cost/price/M3 onto the Item's Costing tab (`custom_tz_costing` , `custom_japan_costing` , etc.).
Cancel / Amend	Submitted	Standard ERPNext cancel/amend if corrections needed after submit.

6.5 How to teach "what do I type?"

- **Fixed rows:** usually leave alone (unless business overrides).
- **Variable rows:** type the car's estimate (TRA Duty from TRA; ICD/Shipping if known per car).
- **Formula rows:** do not type — click Recalculate after changing inputs (Duty).
- **M3 + rates:** keep accurate; they drive many results.

7. Import File Cost Voucher (IFCV) — every field & button

Open: Desk → **Import File Cost Voucher**. One document = one Import File's actual expense allocation worksheet.

7.1 Header fields

Field	What it does
Series	Auto name like `IFCV-2026-00001`.
Import File	**Required.** Selecting it auto-loads Vehicles + Expenses from the database (submitted JE/PI linked to that Import File).
Posting Date	Voucher date (defaults Today).
Auto Update Car Costing Projected	If checked, on **Submit** IFCV writes allocated amounts into matching draft Car Costing lines and recalculates those cars.
Default Distribution Basis	**Equal** or **M3**. Affects how newly loaded By-BL expenses are proposed (By BL Equal vs By BL M3). Changing it reloads from Import File.

7.2 Table: Vehicles on this Import File

Auto-filled list of cars. Columns typically include Item, Chassis, Maker, M3, Stock ID.

- These are the cars that will receive allocations.
- M3 values here drive **By BL M3** math.
- Usually do not invent rows — reload from Import File if cars are missing.

7.3 Table: Posted Expenses (from JE / PI)

Column	Meaning
Expense Account	GL account from JE/PI line.
Description	Prefer JE user remark (often contains chassis) else title.
Amount	Company currency amount to allocate.
Source	JE/PI document name (audit trail).
Costing	Auto if Account matches a Costing; else set manually.
Code	Fetched from Costing.
Distribution Method	By BL Equal · By BL M3 · By Vehicle Manual
Remarks	Optional notes.

> **Stock In Hand rows are ignored on purpose.** Those Purchase Invoice lines are the car purchase stock value (Yen), not a distributeable landed expense.

When you pick a Costing on an expense row:

- If head Allocation Scope = By Vehicle → method becomes **By Vehicle Manual**.
- If head is By BL → method becomes Equal or M3 based on Default Distribution Basis.

7.4 Table: Mapped Vehicle Allocations

Result table: each row is “this Item gets this Costing / account for this amount, from expense row #N”.

- Auto-built on Save for Equal / M3.
- For Manual: filled by dialog (or auto if chassis found in description/remark).
- **Total Expense Amount** should equal **Total Allocated Amount** before submit.

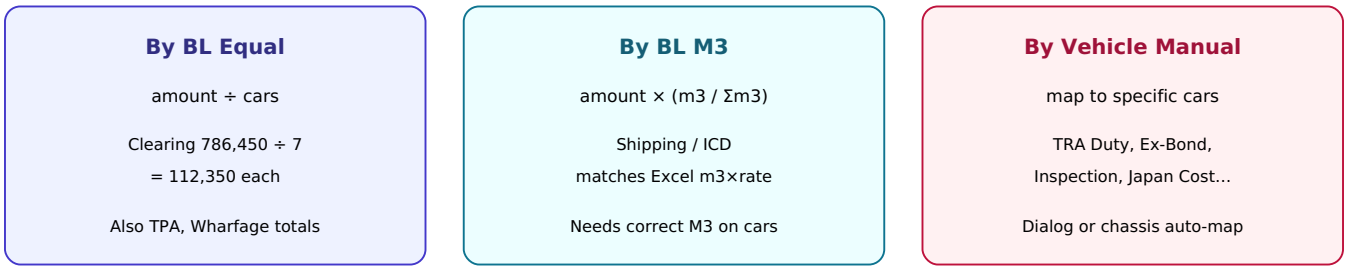


Figure C — The three distribution methods you must teach.

7.5 Buttons on IFCV

Control	What it does	When to use
Actions → Reload from Import File	Re-reads vehicles + JE/PI expenses for the selected Import File. Clears and rebuilds Vehicles/Expenses tables (allocations cleared until save).	After new JVs/PIs were posted, or if tables look stale.
Actions → Rebuild Allocations	Simply **saves** the document, which runs distribution logic again.	After changing distribution methods or amounts.
Actions → Distribute Manually	Opens dialog for expense rows marked By Vehicle Manual. Pick expense row → edit amounts per vehicle → Remaining must be ~0 → Apply. Includes “Quick: put full amount on one car”. If remark contains chassis, amounts default to that car.	Any By Vehicle actual (per-car JE), or when auto-map is wrong.
Save	Validates and rebuilds allocations; updates totals.	Before submit; after edits.
Submit	Locks voucher. If Auto Update is on, pushes allocations into draft Car Costings (matching Import File + Item) and recalculates them.	When allocation review is finished.

7.6 Manual distribution dialog — teach this carefully

1. Save IFCV first (dialog requires a saved name).
2. Actions → **Distribute Manually**.
3. Choose expense row from dropdown (shows code, amount, description/chassis hint).
4. Optional: Quick assign full amount to one Item.
5. Watch summary: Expense | Allocated | Remaining (green when zero).
6. Apply → allocations saved → form reloads.

> **Best practice for Accounts:** When posting a per-car Journal Entry, put the chassis in User Remark (e.g. `ZRR70-0198971 | TOYOTA NOAH…`). IFCV can then auto-map without clicking through every row.

8. Worked example — N.TAN039/26MSC

Import File: N.TAN039/26MSC · **BL:** MEDUW3445865 · **Cars:** 7

1. **Cars already exist as Items** with BL MEDUW3445865 (from Item Excel Import earlier).

2. **Car Costings created (projected)** for all 7 chassis using Excel NEW COST FILE values (M3, TRA Duty, TPA, ICD, Shipping, Fixed defaults). Wharfage calculated as Duty × 7%. Example Noah landed COST = TZS 19,999,903.70.

3. **Actual By BL journals** posted (one total each): Clearing, TPA, Wharfage, ICD, Shipping Line. IFCV splits Equal or M3. ICD/Shipping M3 shares match Excel.

4. **Actual By Vehicle journals** posted per car: TRA Duty, Ex-Bond, Service, Driver, Bond Insurance, Inspection (mapped to Inspection JPY), Japan Cost (mapped to JPN Expense JPY).

5. **IFCV-2026-00001** loads 7 vehicles + expenses, ignores Stock In Hand from the purchase invoice, maps Costing masters via accounts, allocates, balances totals.

6. **Next teaching step:** Submit IFCV with Auto Update on, then open a Car Costing and show the projected lines updated from allocations.

9. FAQ, rules, and common mistakes

Question / mistake	Clear answer
Why is Costing blank on an expense?	JE/PI account is not linked on any Costing. Set Costing.Account to that GL account, then Reload.
Why did Stock In Hand appear before?	It is the car purchase. System now skips Stock / Stock In Hand accounts on purpose.
Why doesn’t Excel COST match our COST?	Excel adds City Levy + VAT from selling price. We show landed COST only (for now).
Wharfage 7% or 7.5%?	Excel formula is Duty×7%. Use 7%. (Some notes sheets said 7.5% — Excel wins.)
Manual dialog says no rows	No expense is set to By Vehicle Manual. Set Costing / method, Save, try again.
Vehicles = 0 on IFCV	Import File BL/supplier invoice doesn’t match Item.custom_bl, or cars disabled.
Recalculate crashed on readiness_html	Fixed: HTML field is display-only; recalculate no longer db_sets it.
Can one BL JV replace seven per-car JVs?	Yes for By BL costs. For By Vehicle actuals, either post per car (best) or post one total and distribute manually in the dialog.

10. Recommended daily workflow checklist

For the Costing / Ops user

1. Confirm Import File and cars exist.
2. Create/open Car Costing per car → Load from Item.
3. Enter Variable estimates (Duty, etc.), confirm M3 & rates.
4. Recalculate → review COST.
5. Keep as Draft until actuals are reviewed (or Submit when estimate is approved).

For the Accounts user

1. Post JE/PI with **Import File** set.
2. Use correct expense accounts (that exist on Costing masters).
3. For per-car costs, put chassis in JE User Remark.
4. Never expect Stock In Hand to be distributed as a costing expense.

For the Costing user after postings

1. Open/create IFCV → select Import File (or Reload).
2. Check Costing masters + distribution methods.
3. Distribute Manually where needed.
4. Confirm Expense total = Allocated total.
5. Submit with Auto Update → spot-check Car Costings.

> **One sentence you can say in a meeting:**
> “Car Costing is our per-car estimate using the Excel rules; Import File Cost Voucher is how we take real journals and invoices on a shipment and fairly attach them to each car — equally, by M3, or by manual/per-car mapping — then push those shares back into the car estimates.”

Document: Car Costing & IFCV Training Manual · vsd_jan_integration · based on live design for Jan International.
Source numbers validated against workspace file `1 BL COST.xlsx` (NEW COST FILE).